Mobile Computing

Uni Gate

2025/2026

Dr. Ahmed Hussien

S. By Eng.Rania

Team Members

SN	Student Name
1	**Rawan Gamal Abdullah Hassanien**
2	**Mariam Mohamed Ahmed Mahmoud**
3	**Shahd Ahmed Mahmoud Ewais**
4	**Shahd Ahmed Ebrahim Ahmed**
5	**Abdelrahman Saad Idrees**
6	**Abdelrahman Samy Abdelhamid**

Project Overview

A comprehensive academic management platform designed to serve students, instructors, and administrators by centralizing academic and administrative services, enabling efficient course and content management, and streamlining communication and academic operations.

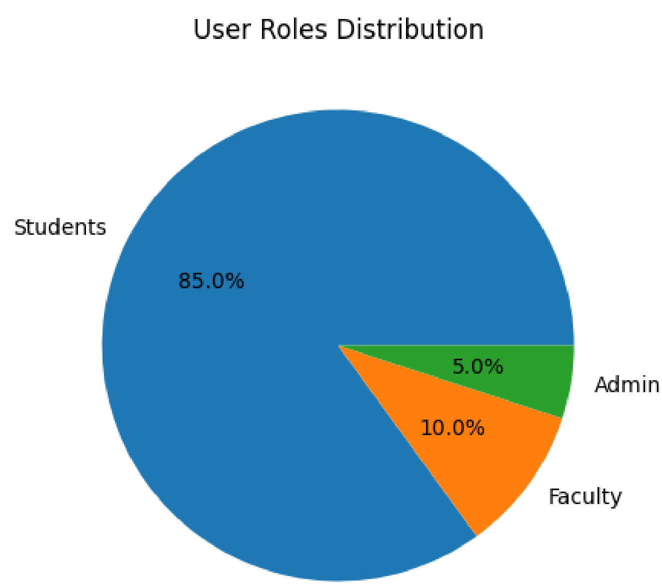
The platform enhances operational efficiency, improves data accuracy, and supports the college's digital transformation strategy.

UX Phase

Stage I: Research

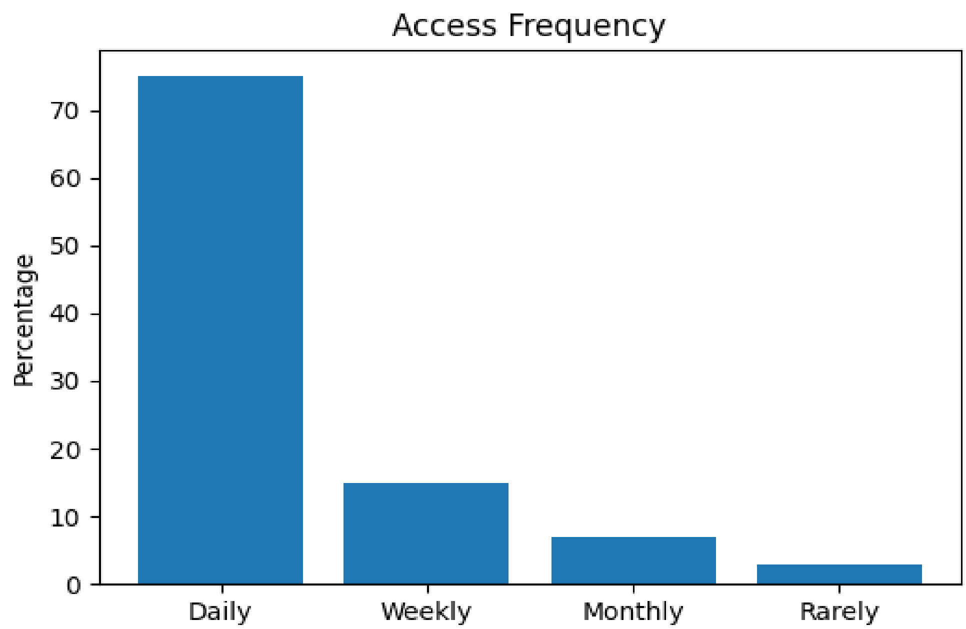
User Demographics

Students represent 85% of the platform users, making them the primary target audience. The system demonstrates strong student engagement and dependency.



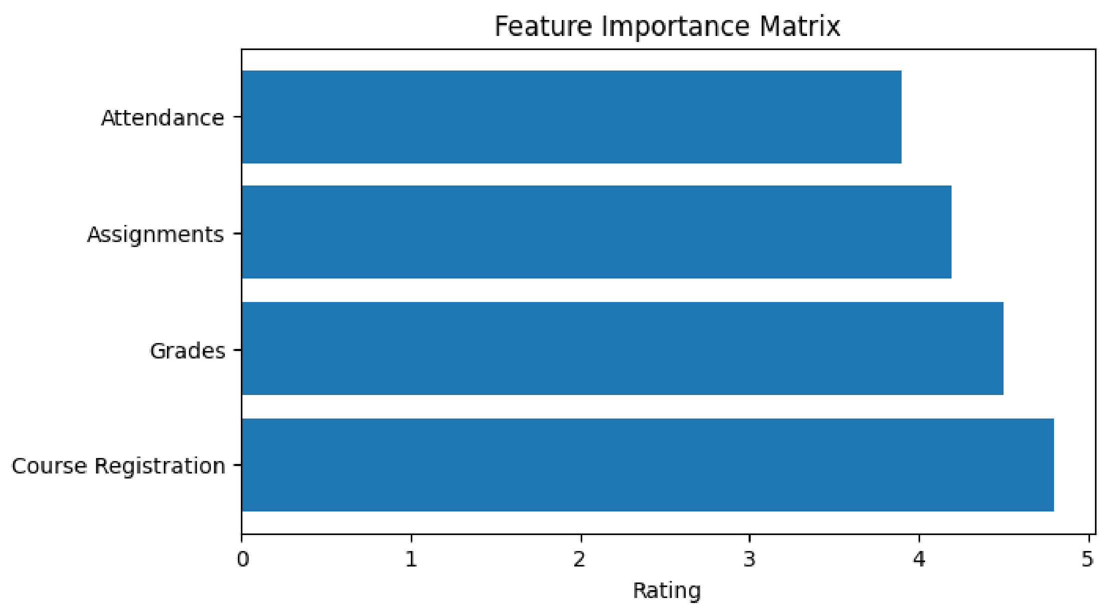
Access Frequency

75% of users access the system daily, confirming successful adoption and operational importance.



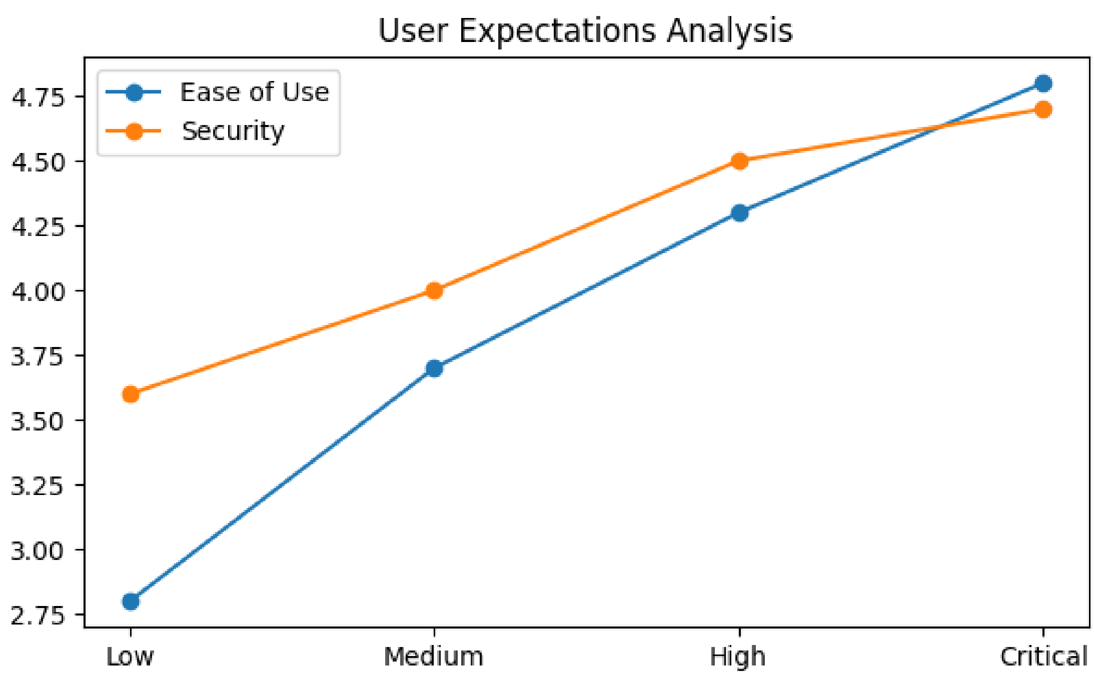
Feature Importance

Course Registration and Grade Viewing received the highest ratings, emphasizing academic workflow efficiency.



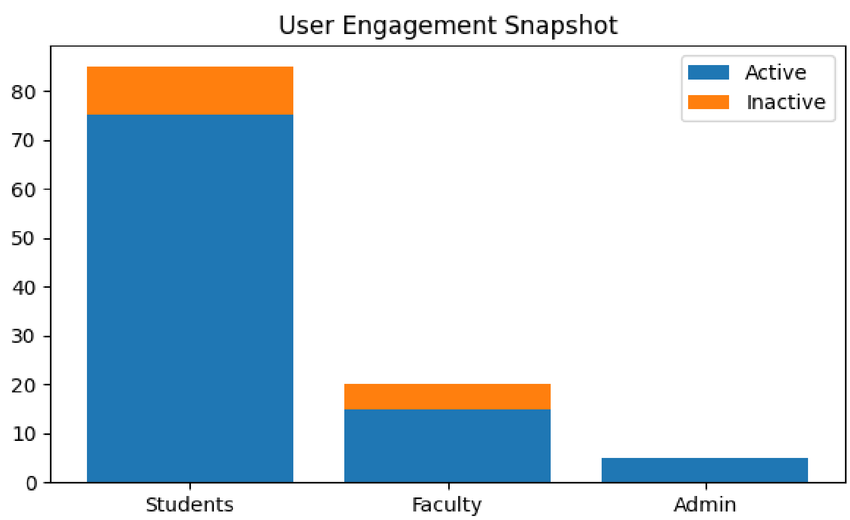
User Expectations

Security and Ease of Use remain the highest priorities across all user groups.



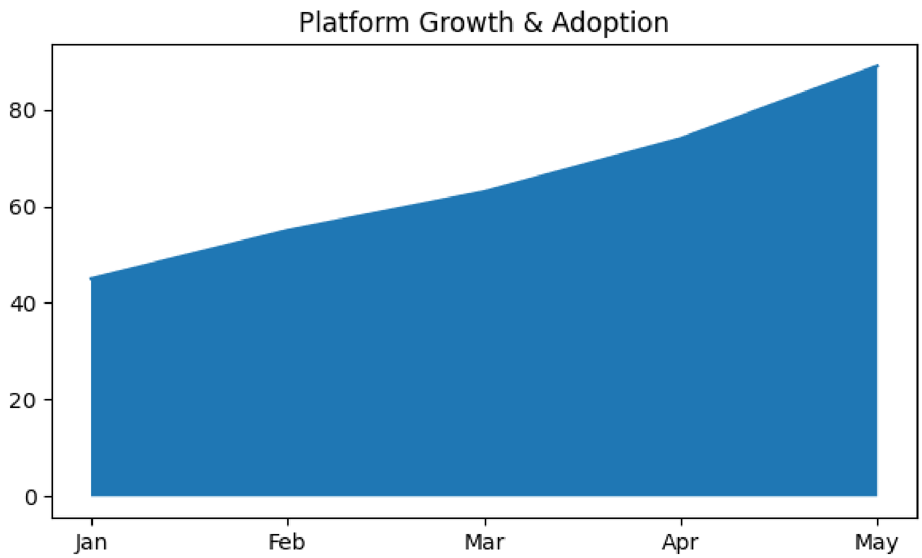
Engagement Snapshot

Students demonstrate the highest activity levels while faculty and admin participation remains moderate.



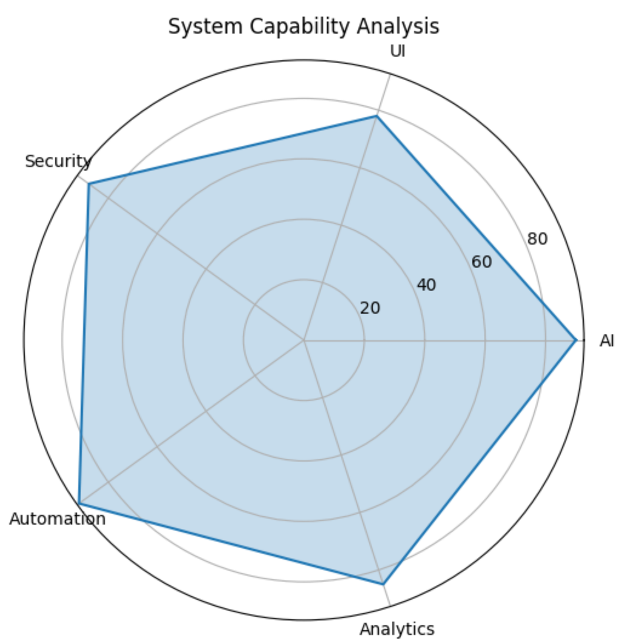
Platform Growth

Continuous increase in adoption rates indicates positive acceptance of digital transformation.



System Capability Analysis

AI integration and automation achieved the strongest evaluation scores among proposed features.

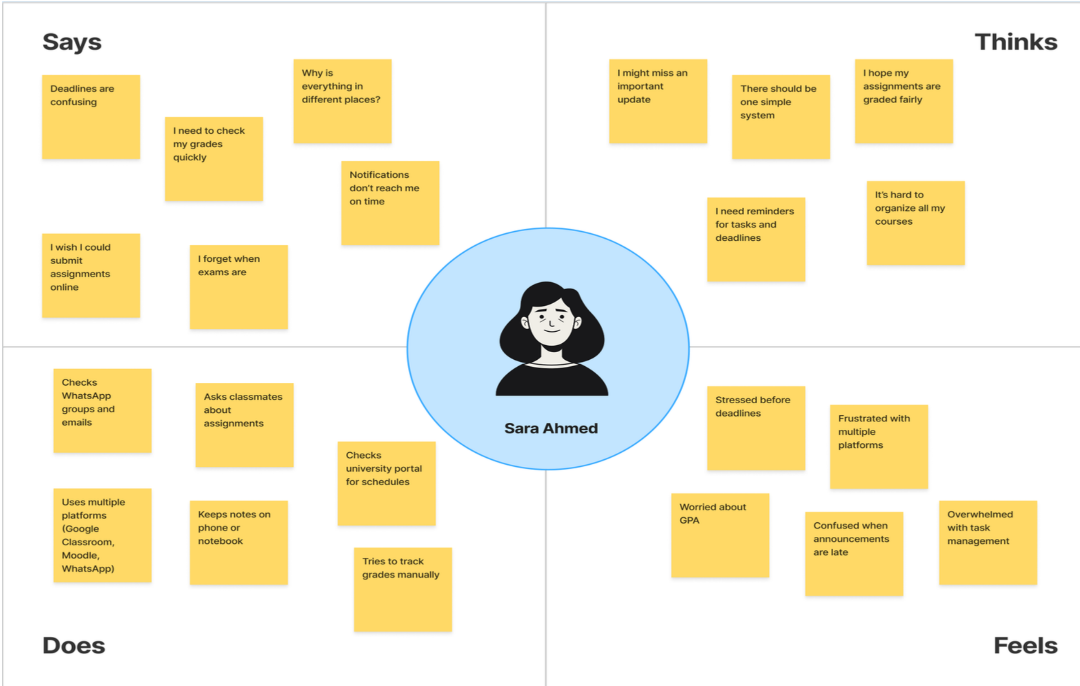


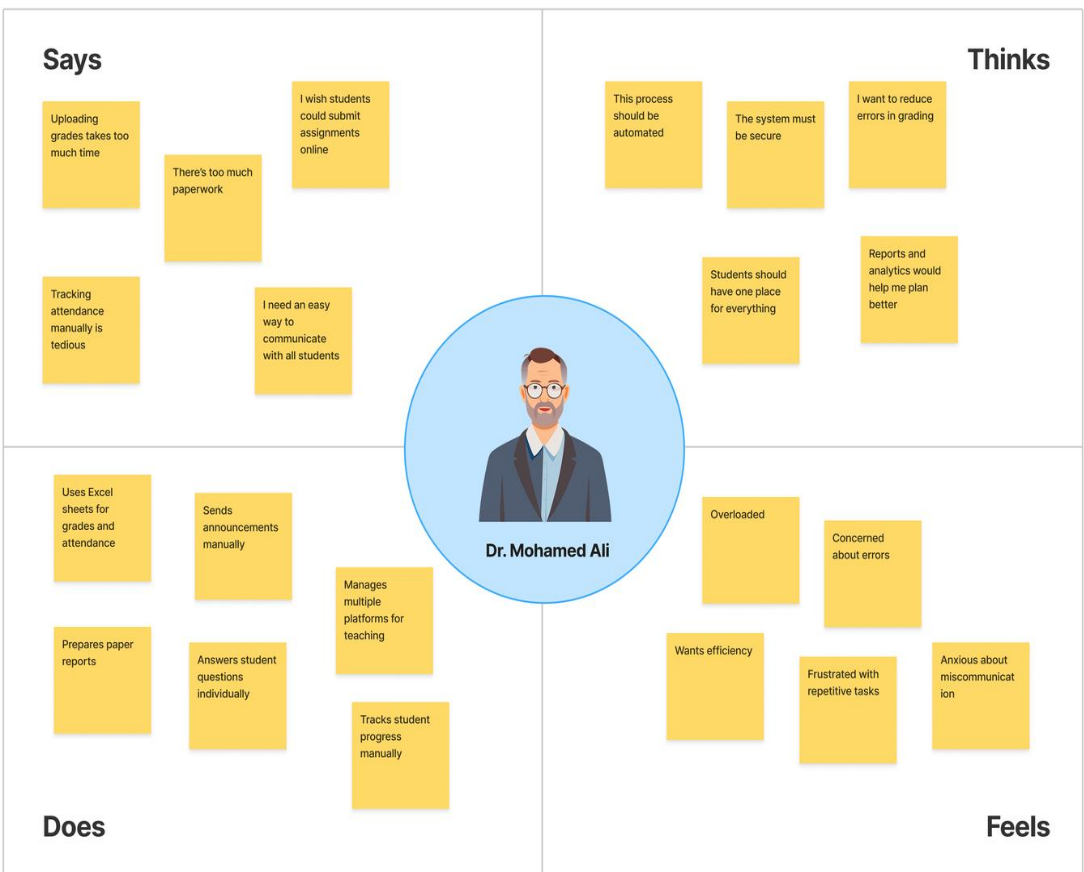
Student Feature Statistics

Feature	Importance %	Satisfaction	Pain Point	Gain
Course Registration	92%	Very High	Peak-time overload	Fast enrollment
Assignments	88%	High	Deadline tracking	Easy submissions
AI Chatbot	84%	High	Response accuracy	24/7 support
Lecture Transcription	81%	High	Language quality	Accessibility
Dashboard	90%	Very High	Complex widgets	Centralized tracking
Attendance	78%	Moderate	QR issues	Automation

Gain and Pain Points

Gains	Pain Points
Automation of academic workflows	UI complexity for new users
Faster communication	Data silos between modules
AI-assisted learning	Limited mobile optimization
Reduced paperwork	Learning curve for advanced tools


Says
Deadlines are confusing
I need to check my grades quickly
Why is everything in different places?
Notifications don't reach me on time
I wish I could submit assignments online
I forget when exams are
Thinks
I might miss an important update
There should be one simple system
I hope my assignments are graded fairly
I need reminders for tasks and deadlines
It's hard to organize all my courses
Sara Ahmed
Checks WhatsApp groups and emails
Asks classmates about assignments
Checks university portal for schedules
Uses multiple platforms (Google Classroom, Moodle, WhatsApp)
Keeps notes on phone or notebook
Tries to track grades manually
Does
Stressed before deadlines
Frustrated with multiple platforms
Worried about GPA
Confused when announcements are late
Overwhelmed with task management
Feels


Says
Uploading grades takes too much time
There's too much paperwork
I wish students could submit assignments online
Tracking attendance manually is tedious
I need an easy way to communicate with all students
Thinks
This process should be automated
The system must be secure
I want to reduce errors in grading
Students should have one place for everything
Reports and analytics would help me plan better
Dr. Mohamed Ali
Uses Excel sheets for grades and attendance
Sends announcements manually
Manages multiple platforms for teaching
Prepares paper reports
Answers student questions individually
Tracks student progress manually
Does
Overloaded
Concerned about errors
Wants efficiency
Frustrated with repetitive tasks
Anxious about miscommunicat ion
Feels

"I just want everything in one organized system."

AGE	21
Role	Third-Year Student
MAJOR	Computer Engineering

PASSIONATE · EMPATHETIC · CURIOUS · ADVENTUROUS · Friendly · Hardworking · Confident · Responsible

USER PERSONA

Sara Ahmed

ABOUT

Sara is a hardworking university student who manages multiple courses each semester. She relies heavily on her phone to check announcements, communicate with classmates, and access academic materials. She prefers fast and simple digital solutions that save her time.

She balances studying, projects, and sometimes extracurricular activities, so organization is very important to her.

GOALS

- View grades quickly
- Access assignments easily
- Avoid missing deadlinesengineers
- Track academic progress

NEEDS

- Simple dashboard
- Notifications
- Clear grade tracking
- Mobile-friendly system

FRUSTRATIONS

- Deadlines are unclear
- Uses multiple platforms (WhatsApp, Email)
- No centralized system
- Worries about missing updates

PERSONALITY

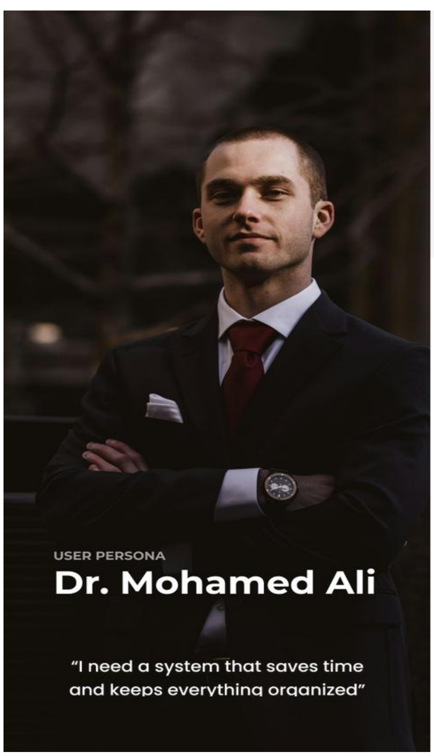
Focused — Distracted

Curious — Passive

Independent — Collaborative

Structured — Flexible

Proactive — Reactive



ABOUT

Dr. Mohamed is a university lecturer responsible for managing courses, assignments, and student grades. He prefers organized digital tools that reduce manual work and save time.

AGE	42
ROLE	University Professor
MAJOR	Computer Engineering

NEEDS

- Simple upload system
- Secure grade submission
- Fast and reliable platform
- Clear student performance overview

FRUSTRATIONS

- Manual grade entry takes time
- Paper-based reports
- Repeated administrative tasks
- Concern about data errors

GOALS

- Upload lectures and materials easily
- Submit grades digitally
- Track student performance
- Reduce paperwork

PERSONALITY

PASSIONATE · MOTIVATIONAL · GIVING · LOVING · OPTIMISTIC · Supportive · Experienced · Helpful · Organized · Good communicator

Stage II: Define a problem

Pain Points & Categories

The following pain points were identified across all user types in the system:

Students

- Difficulty tracking assignments, grades, and attendance across fragmented tools
- No centralized access to lecture materials, videos, and course resources
- Limited support for self-study outside classroom hours
- Lack of visibility into financial status and payment history

Instructors

- Time-consuming manual creation of assessments
- No automated way to transcribe or extract content from lecture videos
- Limited tools for monitoring student engagement and submission status

Admins

- No unified dashboard to manage users, courses, and approval workflows
- Difficulty detecting system misuse or monitoring platform performance
- Manual communication across departments causing delays

5Ws and H Analysis

Who	University students, instructors, and administrators
What	Lack of a unified, AI-integrated academic management platform
Where	University environment (on-campus & remote)
When	During registration, lectures, assignments, and payments
Why	Fragmented tools cause inefficiency, miscommunication, and poor academic tracking
How	By building an all-in-one LMS with AI-powered features, automated workflows, and real-time communication

Problem Statements

Problem Statement 1 — Student

Problem Statement

Ahmed is a/an university student **who needs** a centralized way to track assignments, grades, and attendance in real time **because** scattered systems and delayed updates make it difficult to stay organized, avoid missing deadlines, or prevent failing due to low attendance.

Problem Statement 2 — Instructor

Problem Statement

Dr. Khaled is a/an university instructor **who needs** an efficient way to create lecture materials, generate assessments, and monitor student performance **because** manually preparing quizzes, tracking submissions, and grading across large classes consumes excessive time and reduces teaching quality.

Problem Statement 3 — Admin

Problem Statement

The Academic Administrator is a/an university system manager **who needs** a reliable, centralized dashboard to manage users, courses, approvals, and communications **because** handling these tasks across disconnected systems leads to errors, delays, and inability to monitor platform integrity effectively.

Stage III: Ideate a solution

Goal Statements

Goal Statement 1 — Student

Goal Statement

Our unified student dashboard **will let users** view grades, attendance, assignments, and schedules all in one place, **which will affect** all enrolled university students, **by** reducing confusion, missed deadlines, and academic stress. **We will measure effectiveness by** tracking reduction in missed submissions, GPA improvement, and decreased academic support tickets.

Goal Statement 3 — Instructor

Goal Statement

Our course and assignment management portal **will let users** create assignments, set deadlines, upload learning materials, and track student submissions in one place, **which will affect** instructors managing multiple courses and large numbers of students, **by** reducing administrative overhead, improving organization, and ensuring timely feedback delivery. **We will measure effectiveness by** tracking reduction in missed grading deadlines, instructor time saved per course, and student satisfaction with feedback quality.

Goal Statement 3 — Admin

Goal Statement

Our centralized admin management system **will let users** approve registrations, manage users/courses, and monitor platform activity in real time, **which will affect** all system stakeholders (students, instructors, departments), **by** ensuring smooth operations, policy enforcement, and data integrity. **We will measure effectiveness by** tracking approval response time, reduction in system errors, and user satisfaction scores.

Proposed & Selected Solution

The following solutions were considered during the ideation phase:

#	Solution	Notes
3	Mobile-first student tracker	No instructor/admin side
4 ✓	**Full-stack University Management System**	**SELECTED — covers all user types with AI features**

Reason for Selection:

The full-stack University Management System was selected because it is the only solution that addresses all user pain points simultaneously — students, instructors, and admins .Unlike existing platforms, this system provides measurable improvements in academic outcomes and operational efficiency.

Comparison with Previous Approaches

The following table compares our system against widely-used academic management platforms, highlighting the weak points of existing tools and the features of our system that outperform them:

Weak Points of Existing Systems:

- Limited real-time communication between all user types

- Poor mobile responsiveness and outdated UI/UX design
- No QR-code based attendance tracking

Project Brief Summary

Problem:

University students, instructors, and admins struggle with fragmented digital tools that lack coordination. Current systems suffer from manual assessment creation, poor academic tracking, lack of unified communication channels, and a non-responsive user experience, leading to administrative delays and student confusion.

Selected Solution:

Uni Gate: A comprehensive, unified University Management System featuring three specialized portals (Student, Instructor, and Admin). The system centralizes academic operations—from course registration and payment management to attendance tracking and grading—within a modern, responsive, and user-friendly interface.

Key Proposed Features:

Student Portal:

- Integrated Dashboard: Real-time view of GPA, registered courses, pending assignments, weekly schedules, and payment status.
- Academic Management: Seamless course registration, detailed grade tracking, and the ability to download official transcripts in PDF format.
- Attendance System: Secure QR code check-in for lectures with automated attendance statistics and low-attendance alerts.
- Finance & Payments: Online payment support (Visa/Cards) with automated invoice generation and payment history tracking.
- Course Materials: Centralized access to lecture notes, PDFs, and assignment submission links with status tracking.

Instructor Portal:

- Content Management: Easy upload and organization of course materials, lecture notes, and supplementary resources.
- Assessment Tools: Manual creation and management of assignments, setting deadlines, and recording grades.
- Performance Analytics: Detailed tracking of student submissions and performance, with features to export grade reports and attendance sheets.
- Communication Hub: Direct messaging with students, broadcasting urgent announcements, and hosting discussion boards or virtual office hours.

Admin Portal:

- User & Role Management: Full control over adding/removing users and assigning specific permissions (Student, Lecturer, Dept. Head, Moderator).

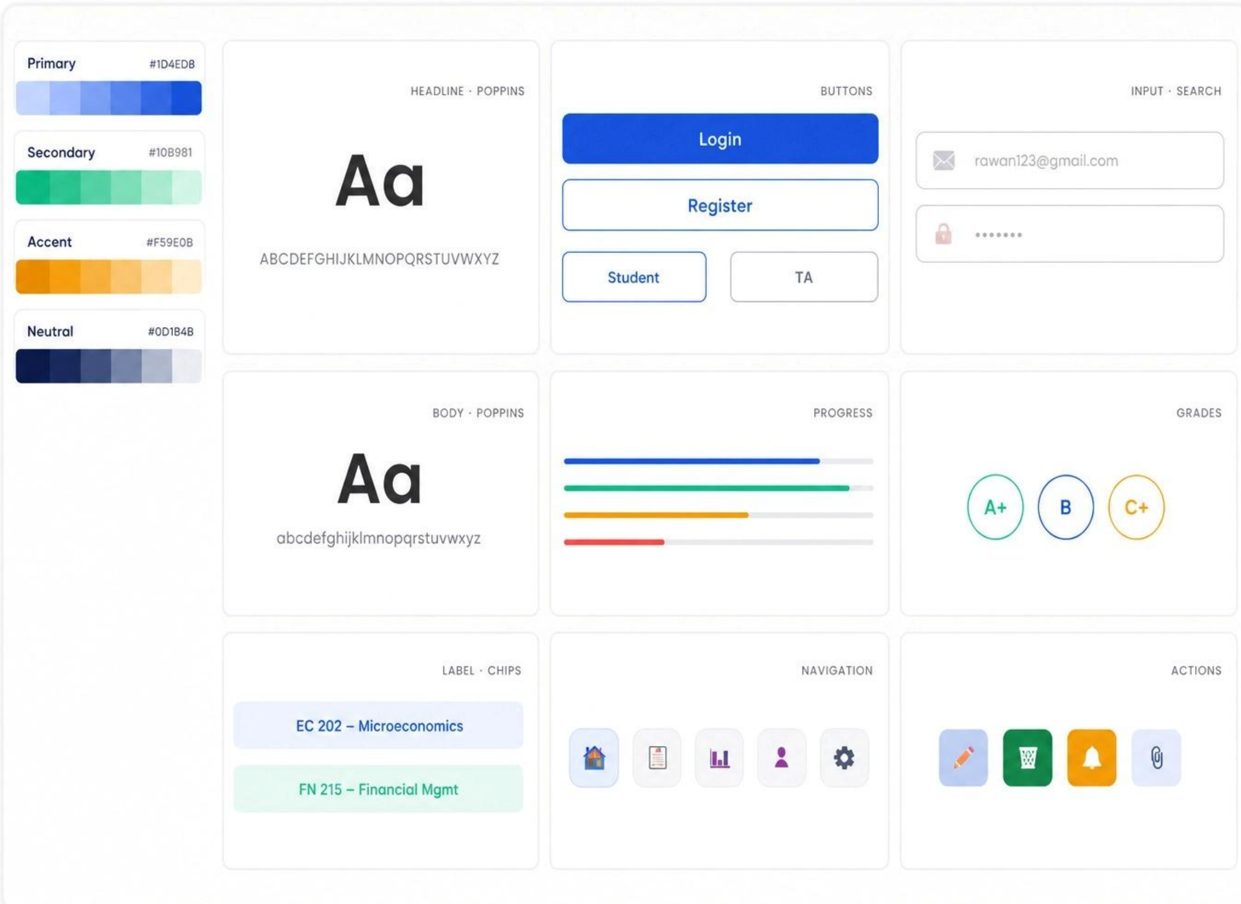
- Academic Infrastructure: Managing departments, majors, course-instructor linking, and class scheduling.
- Approval Workflows: Centralized system for approving course registrations, grade entries, and official student requests (certificates/transcripts).
- Platform Monitoring: Real-time tracking of active users, system activity logs, and resource usage to ensure reliability.
- Broadcasting System: Sending platform-wide notifications, event reminders, and academic deadlines via alerts and emails.

Stage IV

Wireframes & Site map

Frontend

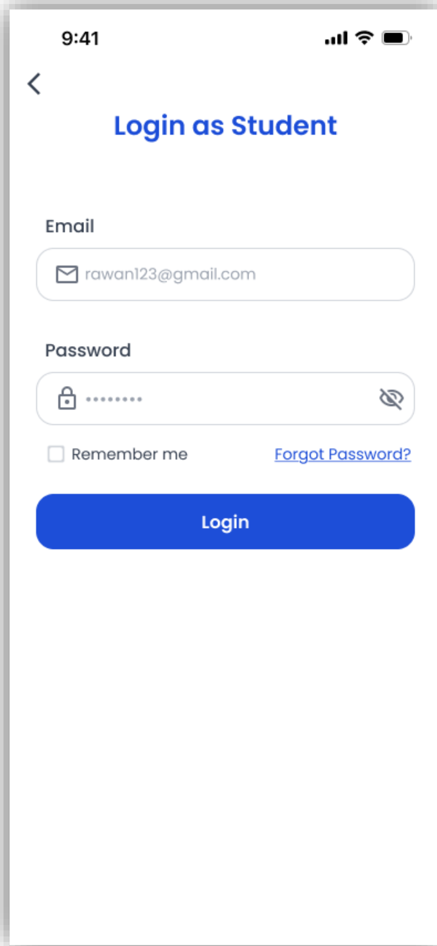
Color Palette & Fonts



UI Design

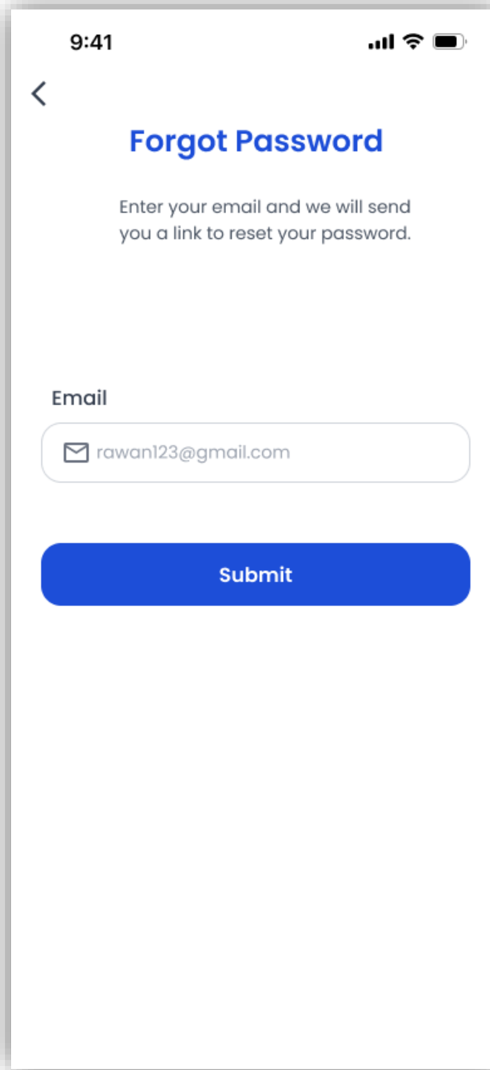
Student Login Page

Description: A secure login interface for students to access their accounts.



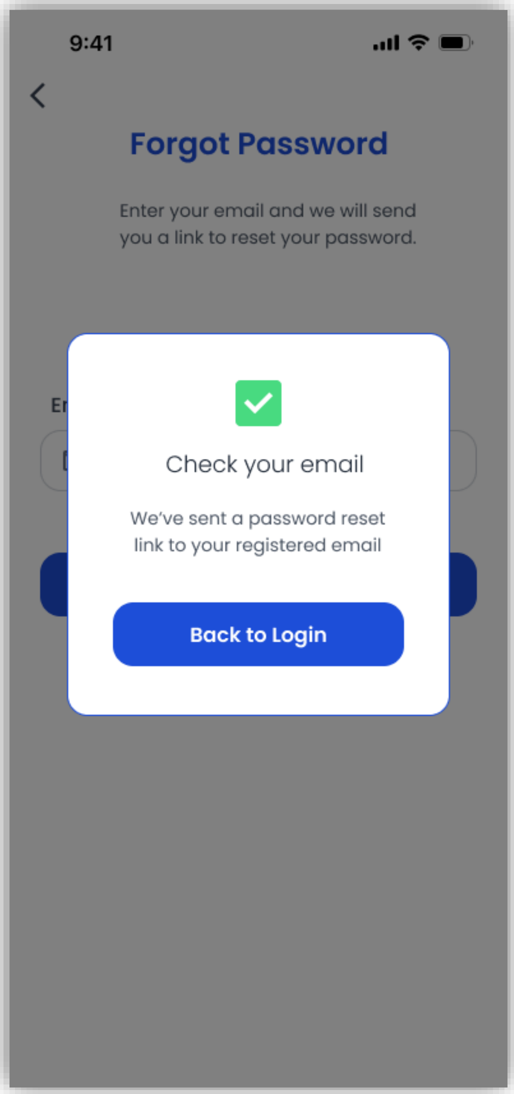
Forgot Password Page

Description: A dedicated screen for users who have lost access to their accounts.



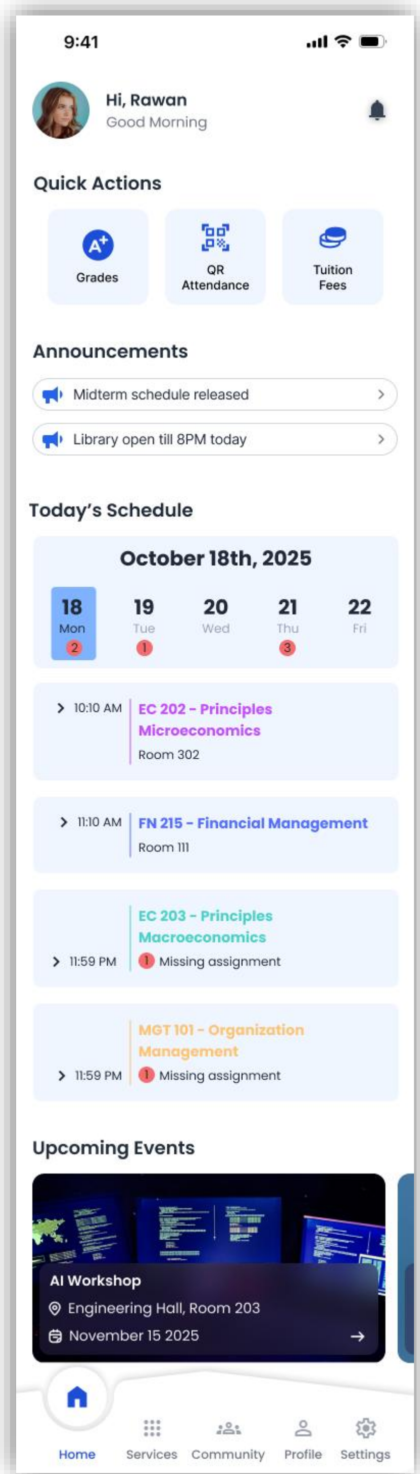
Email Confirmation Pop-up

Description: A confirmation modal that appears after a successful password reset request.



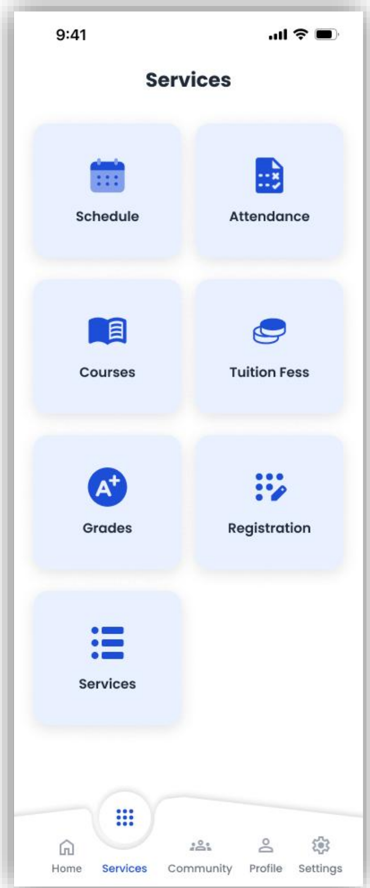
Student Dashboard

Description: The primary landing page providing a personalized overview of the student's academic day.



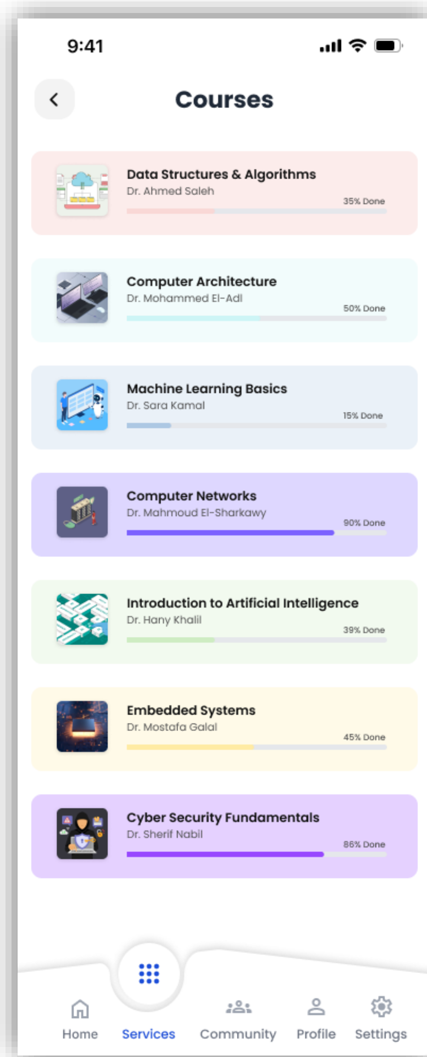
Services Page

Description: A centralized menu for all administrative and academic functions



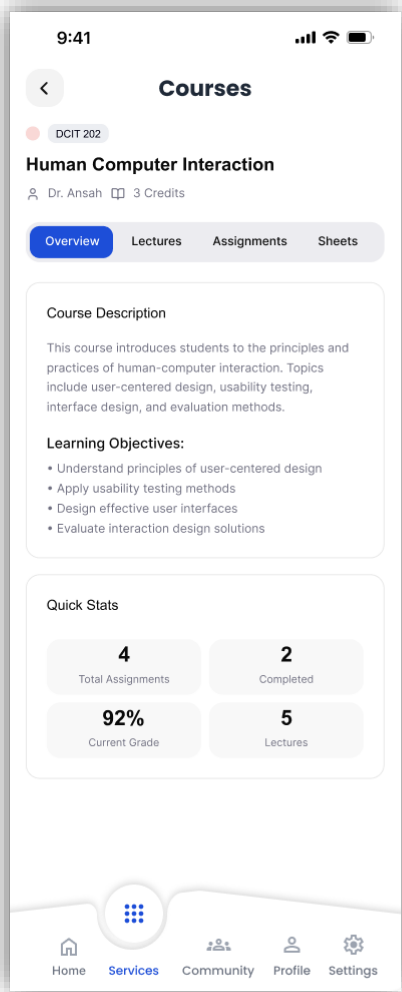
My Courses Page

Description: A summary list of all subjects the student is currently registered for.



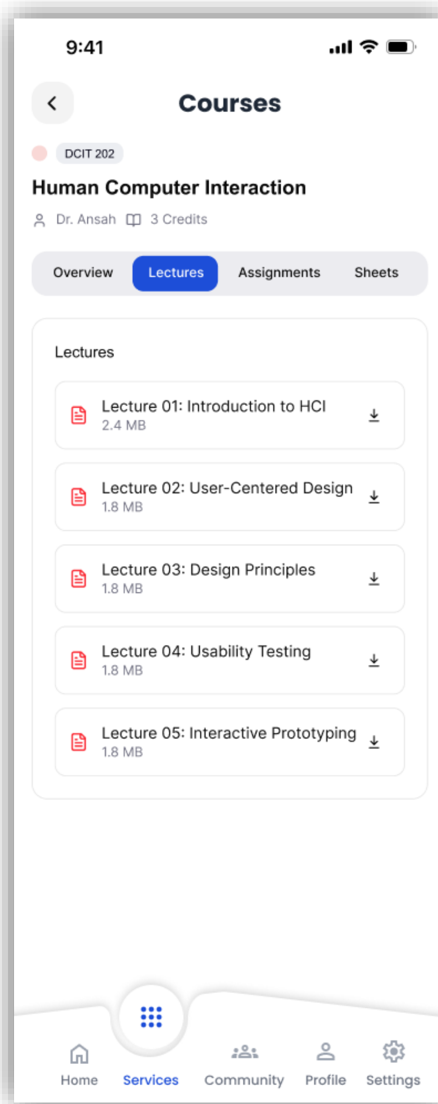
Course Overview Page

Description: A detailed introduction to a specific course, such as "Human Computer Interaction".



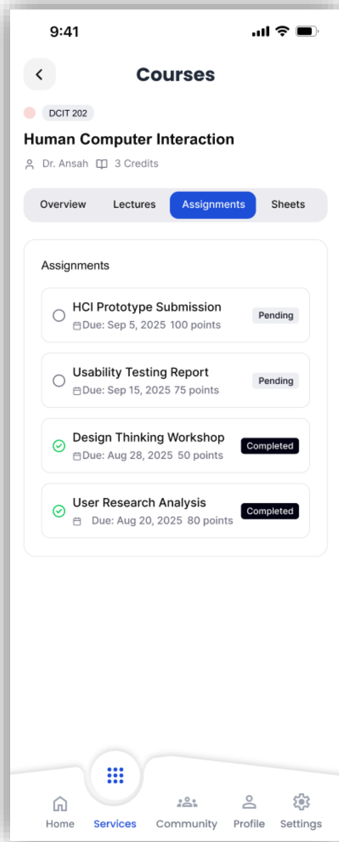
Lectures Page

Description: A repository for all lecture-related documents for a specific course.



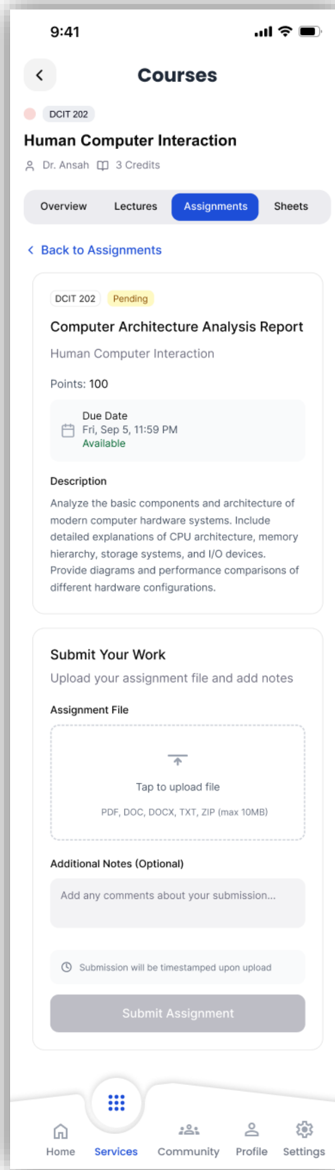
Assignments Page

Description: A management screen for tracking all graded tasks within a course.



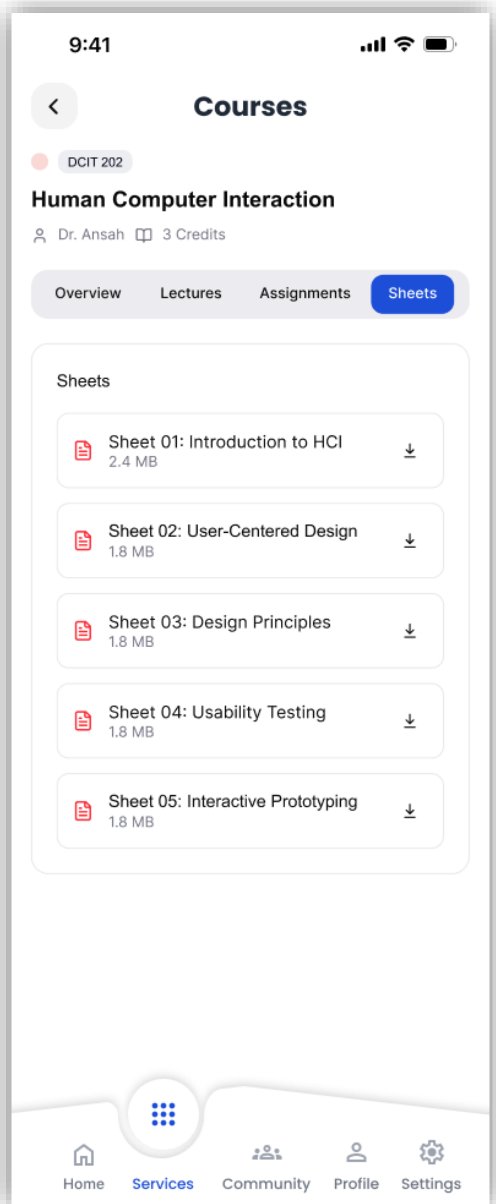
Assignment Submission Page

Description: The interface used to upload completed work for grading.



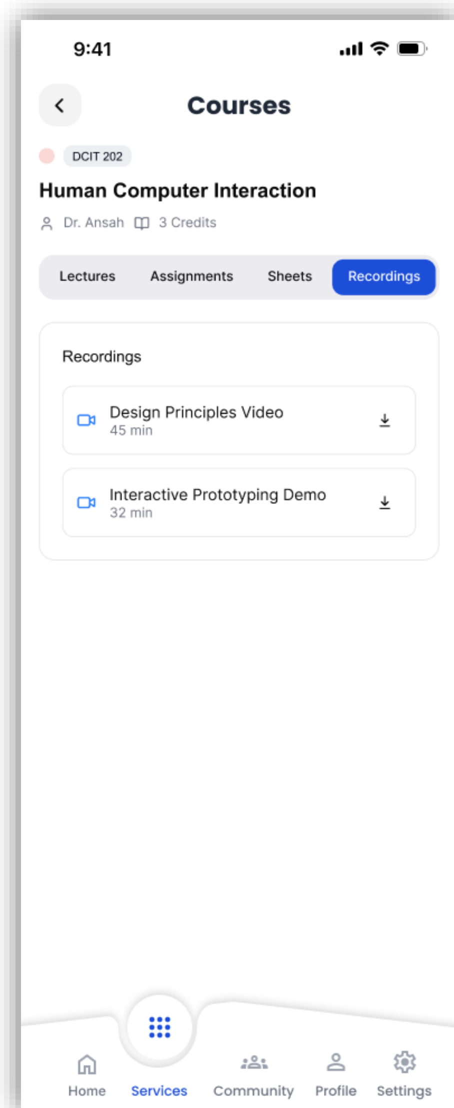
Sheets Page

Description: A section for accessing practice sheets or exercise documents



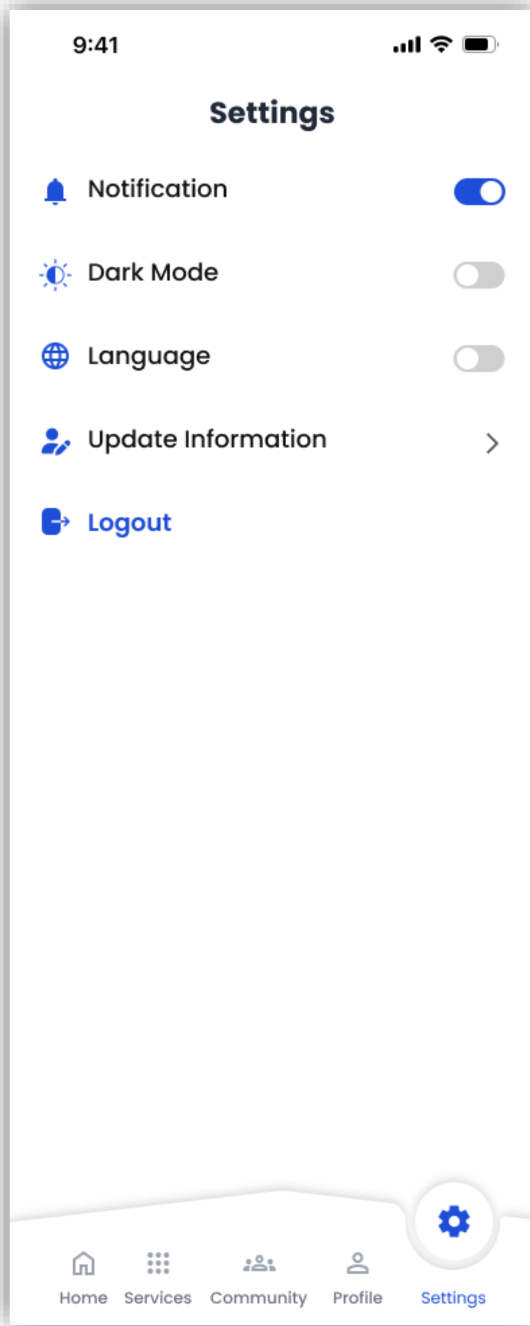
Recordings Page

Description: A gallery of recorded video sessions for student review.

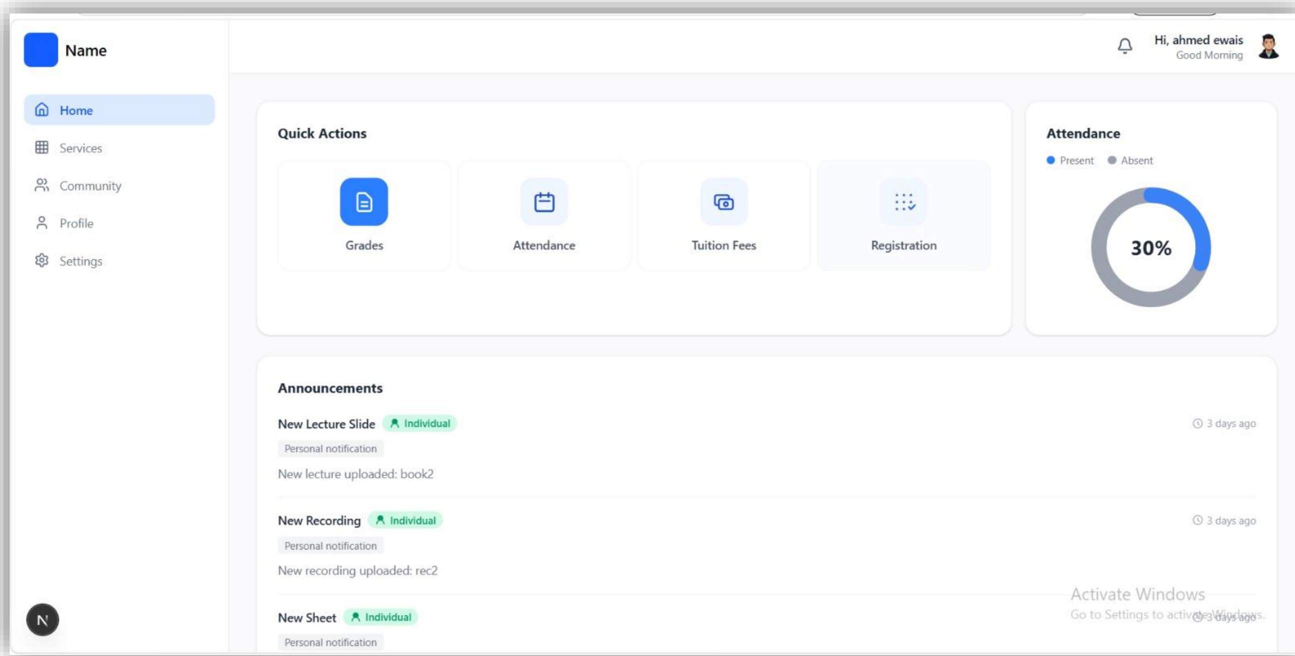


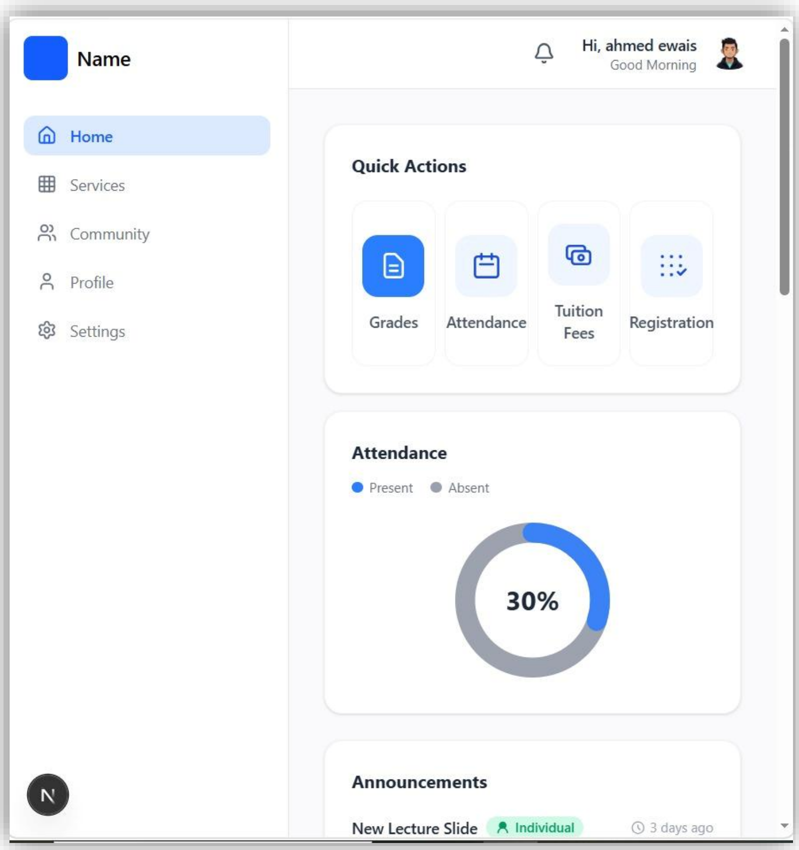
Settings Page

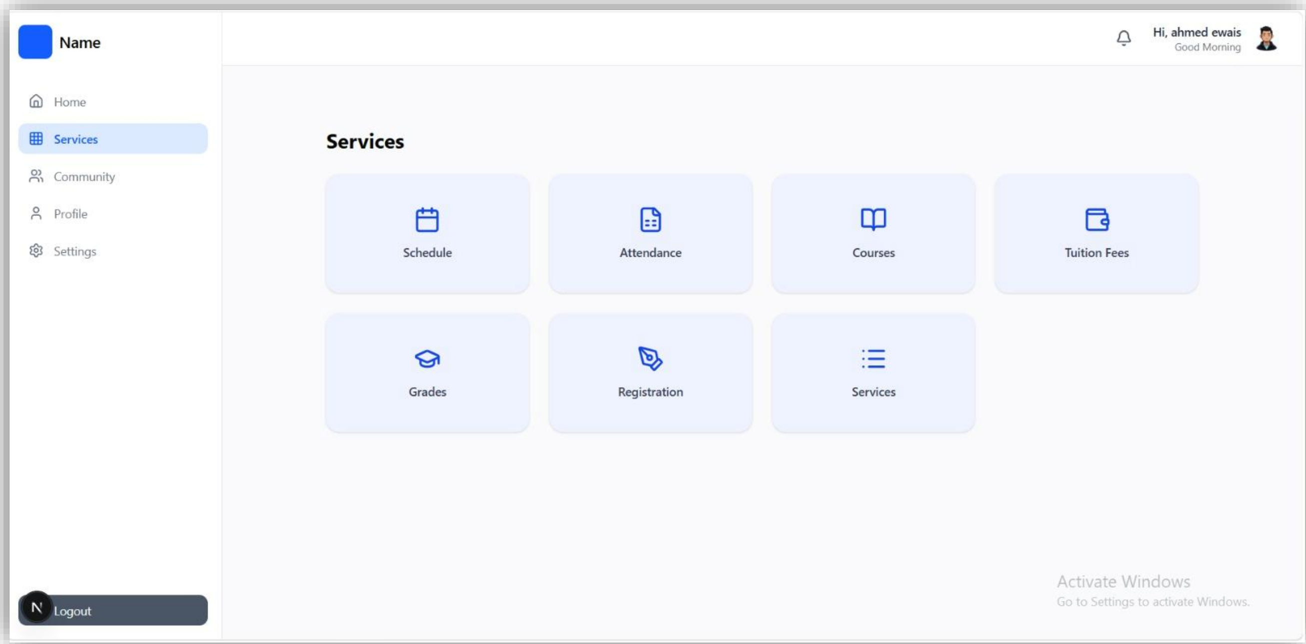
Description: A customization menu to manage the student's app experience.



Responsive Pages






Name
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Good Morning
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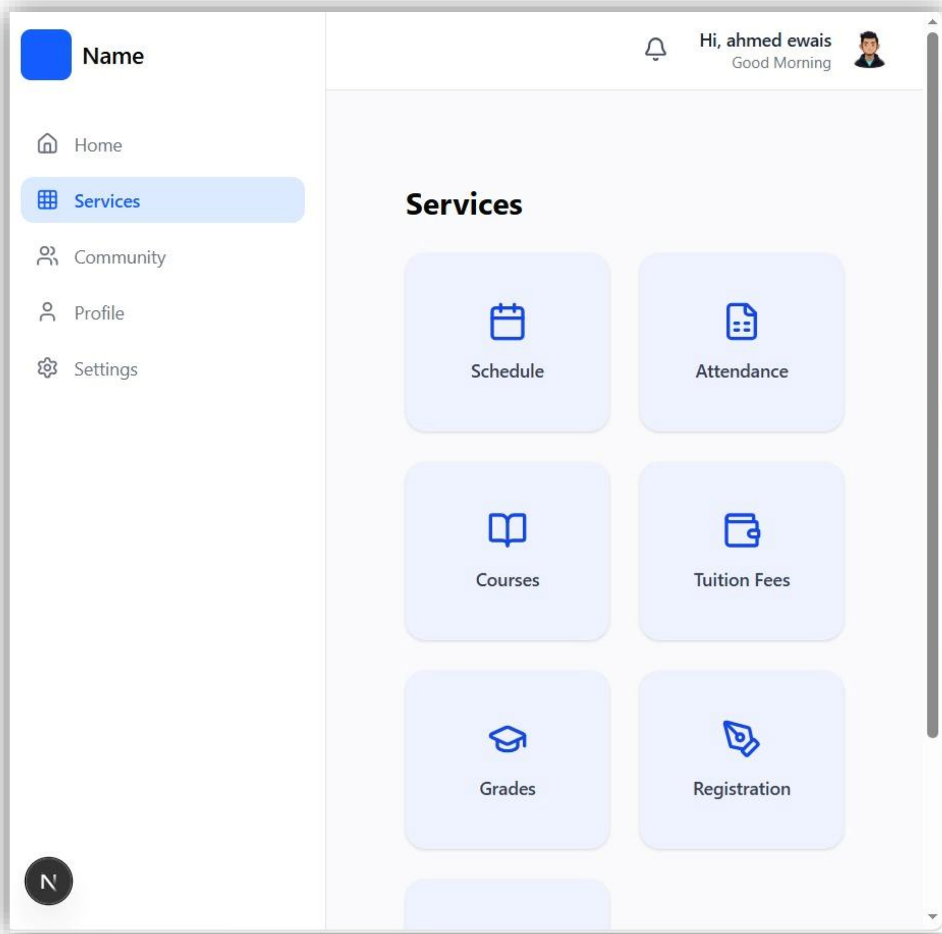
Schedule
Attendance
Courses
Tuition Fees
Grades
Registration
Services
Activate Windows
Go to Settings to activate Windows.


Name
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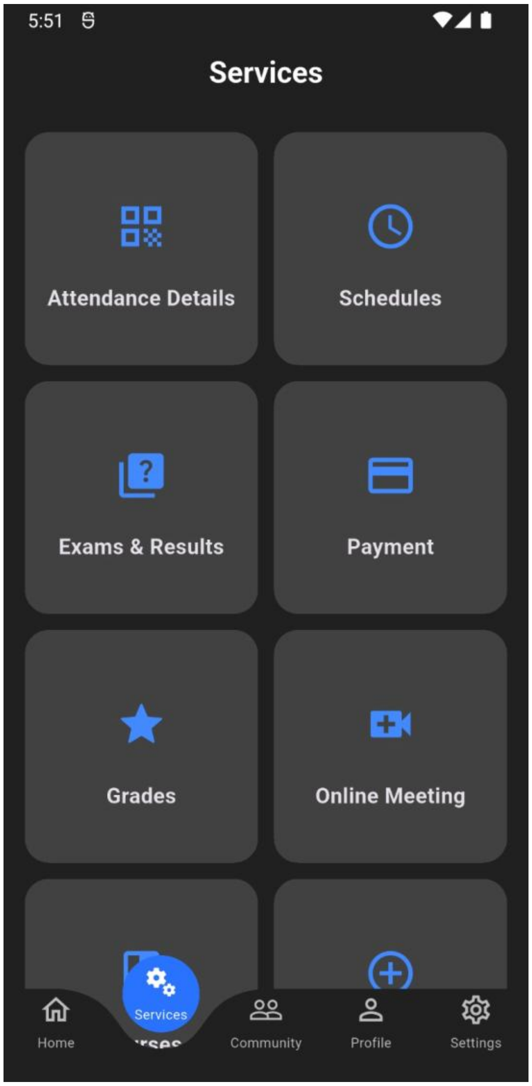
Good Morning
Services
Schedule
Attendance
Courses
Tuition Fees
Grades
Registration

Dark Mode



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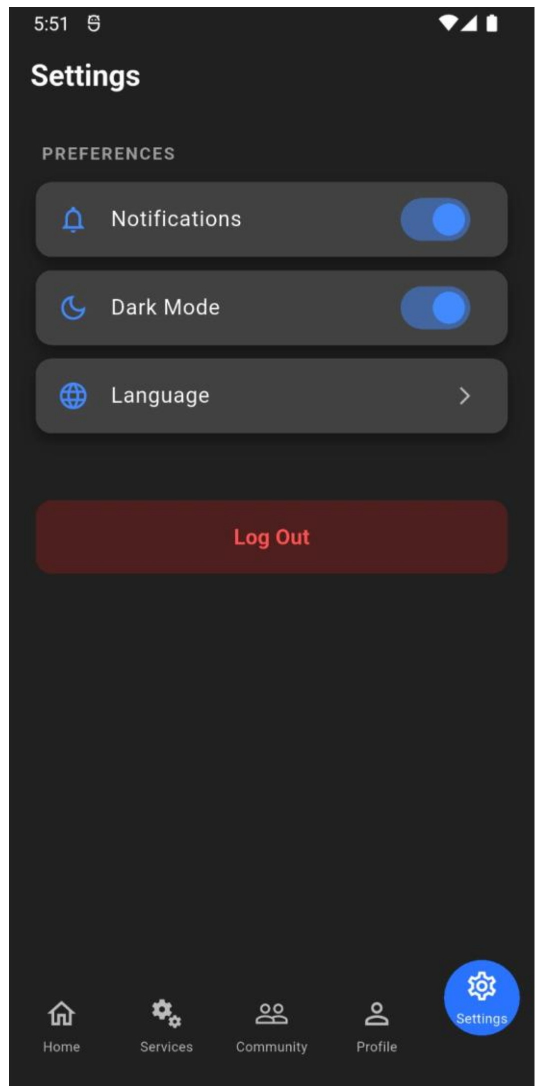
Services
Attendance Details
Schedules
Exams & Results
Payment
Grades
Online Meeting
Home
Services
Community
Profile
Settings



5:51
Settings
PREFERENCES
Notifications
Dark Mode
Language
Log Out
Home
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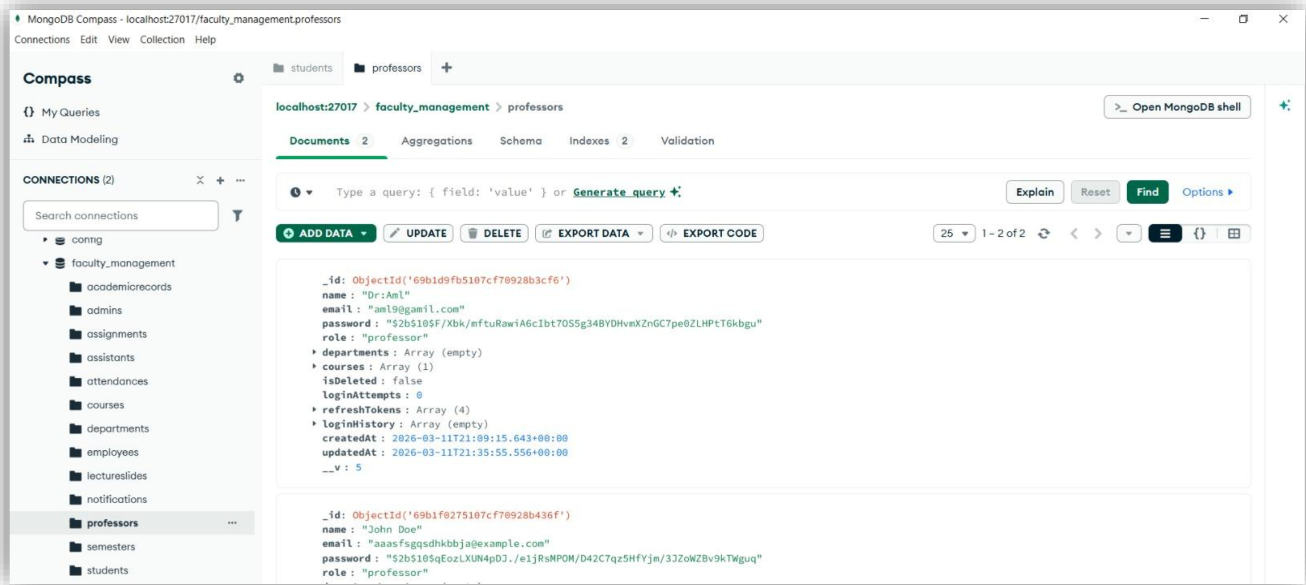
Profile
Settings

Backend



MongoDB Compass - localhost:27017/faculty_management.professors
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (2)
Search connections
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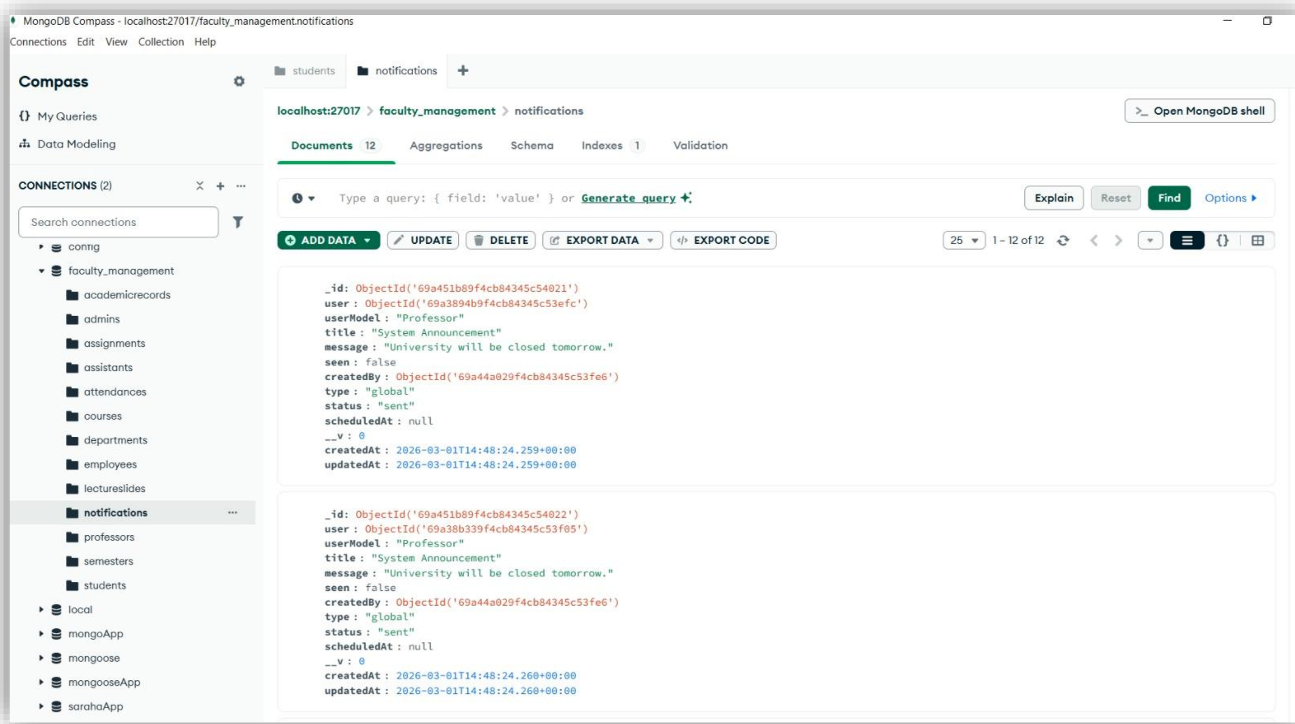
faculty_management
academicrecords
admins
assignments
assistants
attendances
courses
departments
employees
lectureslides
notifications
professors
semesters
students
localhost:27017 > faculty_management > professors
Open MongoDB shell
Documents 2 Aggregations Schema Indexes 2 Validation
Type a query: { field: 'value' } or Generate query
Explain Reset Find Options
ADD DATA UPDATE DELETE EXPORT DATA EXPORT CODE
25 1 - 2 of 2
_id: ObjectId('69b1d9fb5107cf78928b3cf6')
name : "Dr:Aml"
email : "aml9@gamil.com"
role : "professor"
departments : Array (empty)
courses : Array (1)
isDeleted : false
loginAttempts : 0
refreshTokens : Array (4)
loginHistory : Array (empty)
createdAt : 2026-03-11T21:09:15.643+00:00
updatedAt : 2026-03-11T21:35:55.556+00:00
__v : 5
_id: ObjectId('69b1f0275107cf78928b436f')
name : "John Doe"
email : "aaasfsgqsdhkbbja@example.com"
role : "professor"



MongoDB Compass - localhost:27017/faculty_management.notifications
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (2)
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localhost:27017 > faculty_management > notifications
Open MongoDB shell
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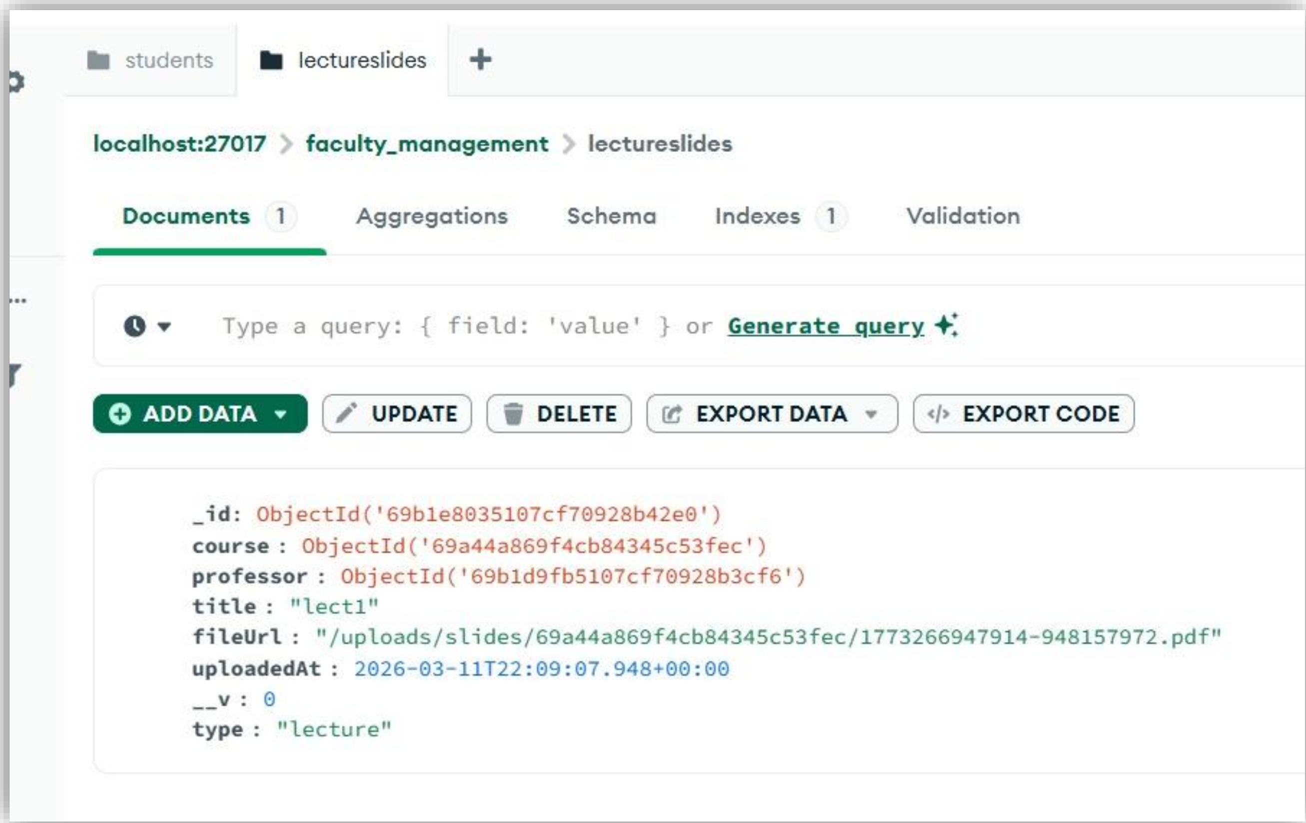
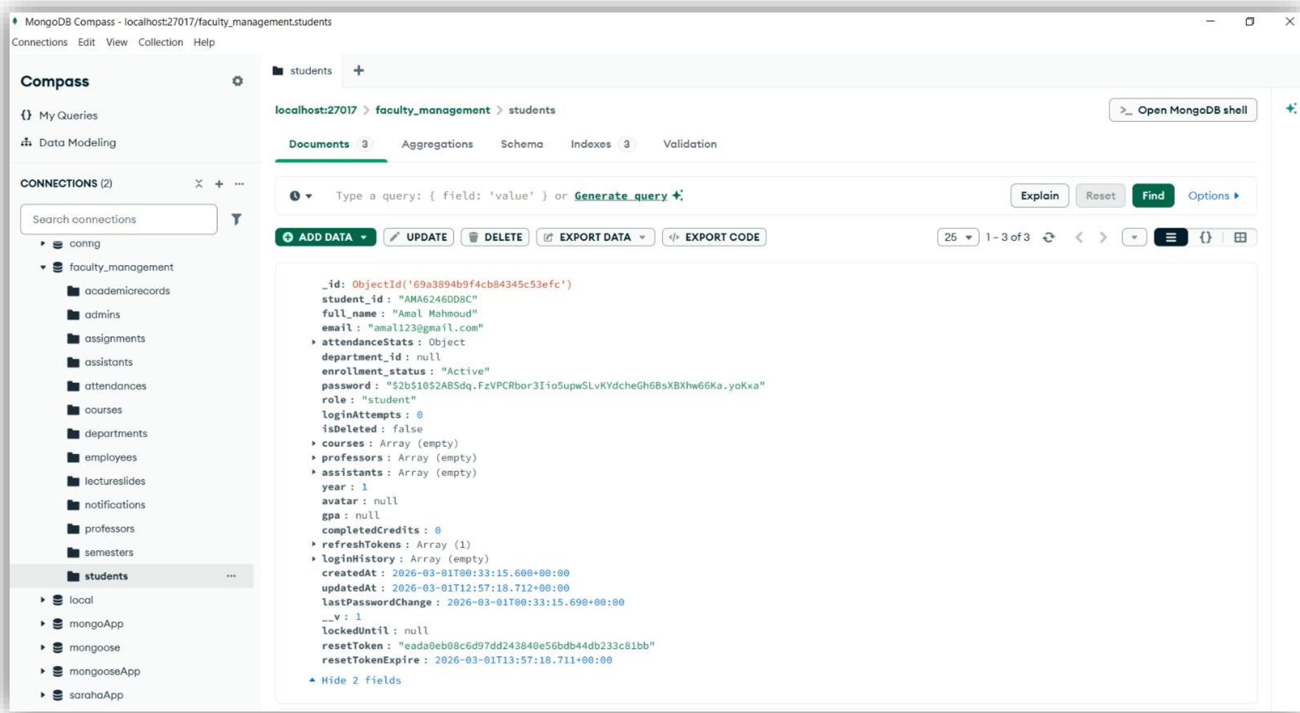
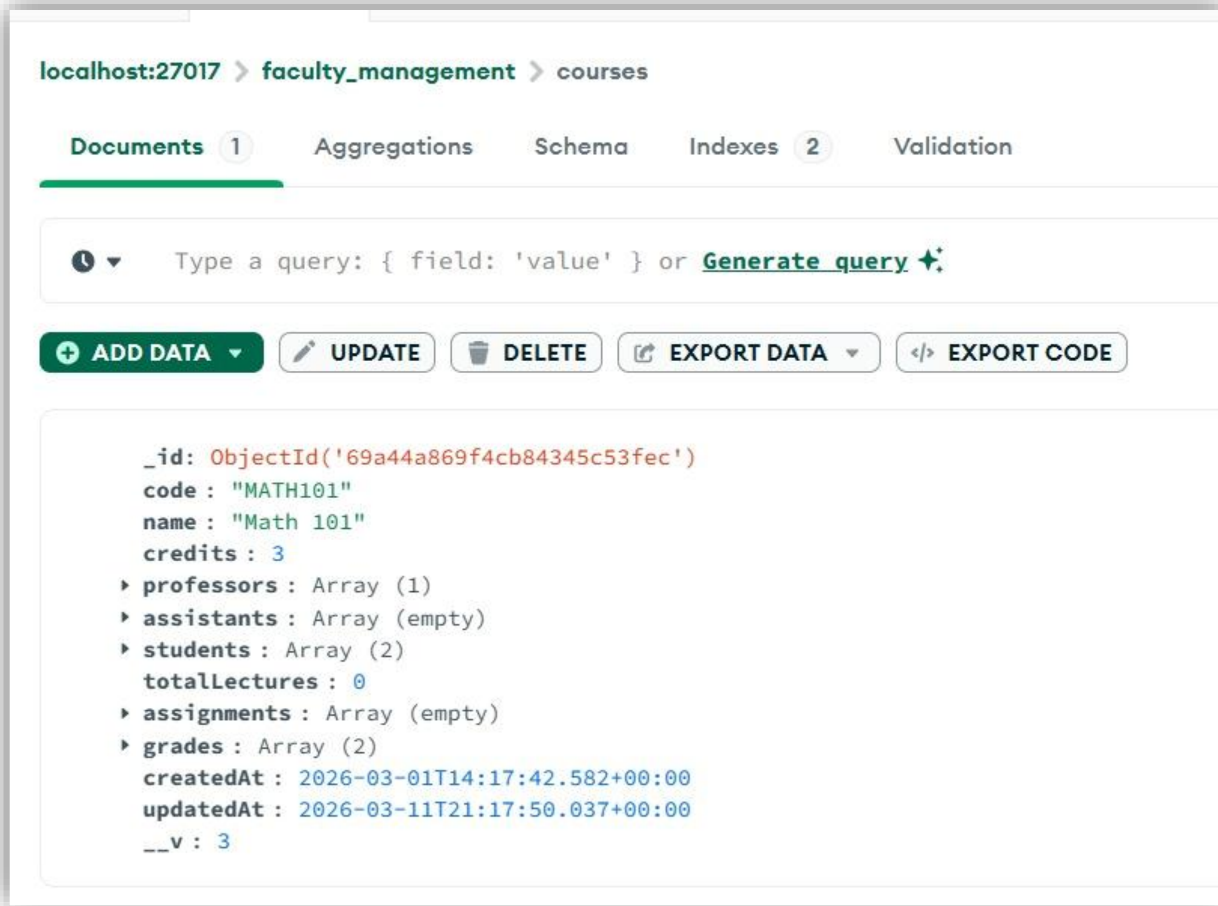
25 1 - 12 of 12
_id: ObjectId('69a451b89f4cb84345c54021')
user : ObjectId('69a3894b9f4cb84345c53efc')
userModel : "Professor"
title : "System Announcement"
message : "University will be closed tomorrow."
seen : false
createdBy : ObjectId('69a44a029f4cb84345c53fe6')
type : "global"
status : "sent"
scheduledAt : null
__v : 0
createdAt : 2026-03-01T14:48:24.259+00:00
updatedAt : 2026-03-01T14:48:24.259+00:00
_id: ObjectId('69a451b89f4cb84345c54022')
user : ObjectId('69a38b339f4cb84345c53f05')
userModel : "Professor"
title : "System Announcement"
message : "University will be closed tomorrow."
seen : false
createdBy : ObjectId('69a44a029f4cb84345c53fe6')
type : "global"
status : "sent"
scheduledAt : null
__v : 0
createdAt : 2026-03-01T14:48:24.260+00:00
updatedAt : 2026-03-01T14:48:24.260+00:00

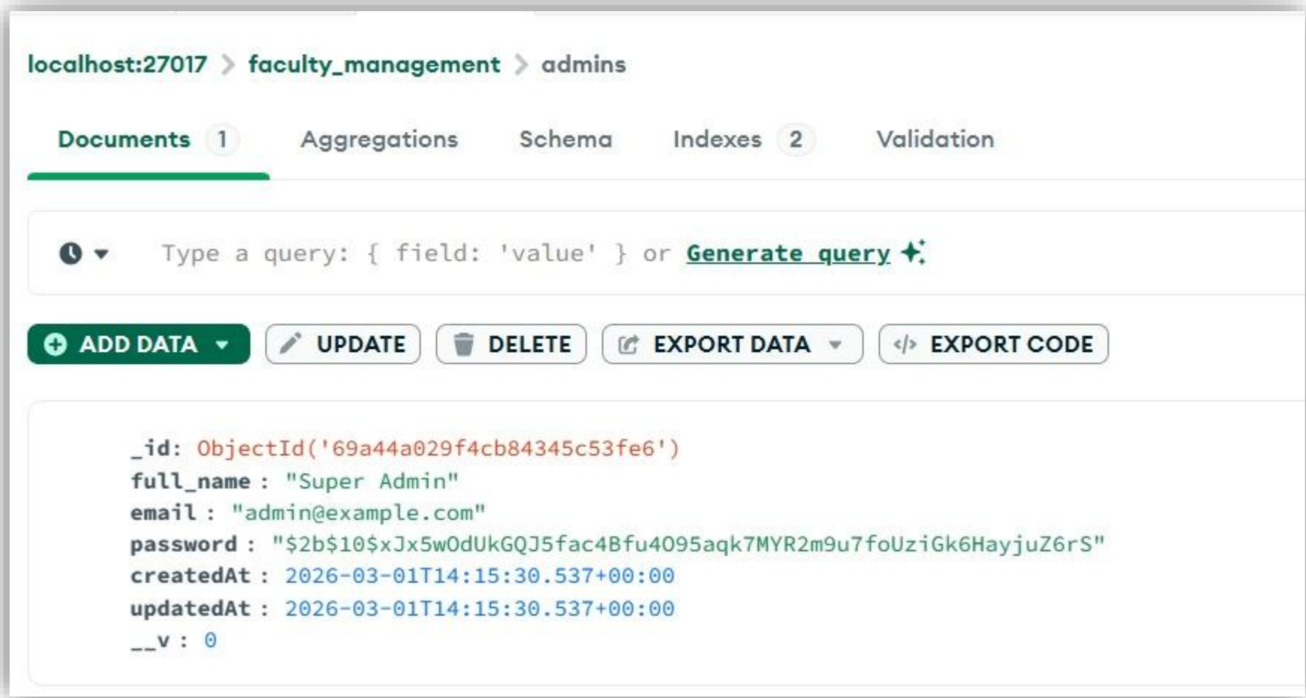


students
lectureslides
localhost:27017 > faculty_management > lectureslides
Documents 1
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE
_id: ObjectId('69b1e8035107cf70928b42e0')
course : ObjectId('69a44a869f4cb84345c53fec')
professor : ObjectId('69b1d9fb5107cf70928b3cf6')
title : "lect1"
fileUrl : "/uploads/slides/69a44a869f4cb84345c53fec/1773266947914-948157972.pdf"
uploadedAt : 2026-03-11T22:09:07.948+00:00
__v : 0
type : "lecture"



MongoDB Compass - localhost:27017/faculty_management.students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (2)
Search connections
config
faculty_management
academicrecords
admins
assignments
assistants
attendances
courses
departments
employees
lectureslides
notifications
professors
semesters
students
local
mongoApp
mongoose
mongooseApp
sarahaApp
students
localhost:27017 > faculty_management > students
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 3
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE
25
1 – 3 of 3
_id: ObjectId('69a3894b9f4cb84345c53efc')
student_id : "AMA6246DD8C"
full_name : "Amal Mahmoud"
email : "amal123@gmail.com"
attendanceStats : Object
department_id : null
enrollment_status : "Active"
password : "$2b$10$2ABSdq.FzVPCRbor3Iio5upwSLvKYdcheGh6BsXBXhw66Ka.yoKxa"
role : "student"
loginAttempts : 0
isDeleted : false
courses : Array (empty)
professors : Array (empty)
assistants : Array (empty)
year : 1
avatar : null
gpa : null
completedCredits : 0
refreshTokens : Array (1)
loginHistory : Array (empty)
createdAt : 2026-03-01T00:33:15.600+00:00
updatedAt : 2026-03-01T12:57:18.712+00:00
lastPasswordChange : 2026-03-01T00:33:15.690+00:00
__v : 1
lockedUntil : null
resetToken : "eada0eb08c6d97dd243840e56bdb44db233c81bb"
resetTokenExpire : 2026-03-01T13:57:18.711+00:00
Hide 2 fields


localhost:27017 > faculty_management > courses
Documents 1
Aggregations
Schema
Indexes 2
Validation
Type a query: { field: 'value' } or Generate query
ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE
_id: ObjectId('69a44a869f4cb84345c53fec')
code : "MATH101"
name : "Math 101"
credits : 3
professors : Array (1)
assistants : Array (empty)
students : Array (2)
totalLectures : 0
assignments : Array (empty)
grades : Array (2)
createdAt : 2026-03-01T14:17:42.582+00:00
updatedAt : 2026-03-11T21:17:50.037+00:00
__v : 3


localhost:27017 > faculty_management > admins
Documents 1
Aggregations
Schema
Indexes 2
Validation
Type a query: { field: 'value' } or Generate query
ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE
_id: ObjectId('69a44a029f4cb84345c53fe6')
full_name : "Super Admin"
email : "admin@example.com"
password : "$2b$10$xJx5wOdUkGQJ5fac4Bfu4095aqk7MYR2m9u7foUziGk6HayjuZ6rS"
createdAt : 2026-03-01T14:15:30.537+00:00
updatedAt : 2026-03-01T14:15:30.537+00:00
__v : 0

Backend Explanation

“The backend of the Student System was built using Node.js and Express.js with MongoDB as the database. The system follows a REST API architecture where the frontend communicates with the backend through API endpoints.”

Database & Dynamic Content

“All system data is stored in MongoDB collections such as:

- Students
- Instructors
- Courses
- Assignments
- Grades
- Submissions

Instead of writing static content inside the pages, the data is fetched dynamically from the database and rendered automatically on the frontend.”

Example of Dynamic Content

Courses Module

“When a student logs in, the frontend sends a request to the backend to retrieve the registered courses from MongoDB.
The backend queries the database and returns the courses as JSON data, then the frontend renders them dynamically.”

Assignment System

“Assignments are also stored in the database with their deadlines and submission status.
Students can upload their solutions, and the backend handles file storage and updates the submission status in MongoDB.”

Grades Module

“Grades are fetched dynamically from the database for each registered course.
The backend sends the student grades through secure API endpoints, and the frontend displays them automatically.”

Layer	Technology	Purpose
Frontend	Next.js	Provides server-side rendering (SSR) and optimized performance for a fast, SEO-friendly interface.
Styling	Tailwind CSS	A utility-first CSS framework used to create the modern, responsive, and clean UI/UX seen in the project screens.
Backend	Node.js	A JavaScript runtime built on Chrome's V8 engine, handling concurrent requests efficiently.
Server Framework	Express.js	A minimal and flexible web application framework for handling API routing and server logic.
Database	MongoDB	A NoSQL database used for flexible and scalable storage of user profiles, course data, and academic records.